Modelling culture formation on the European map*

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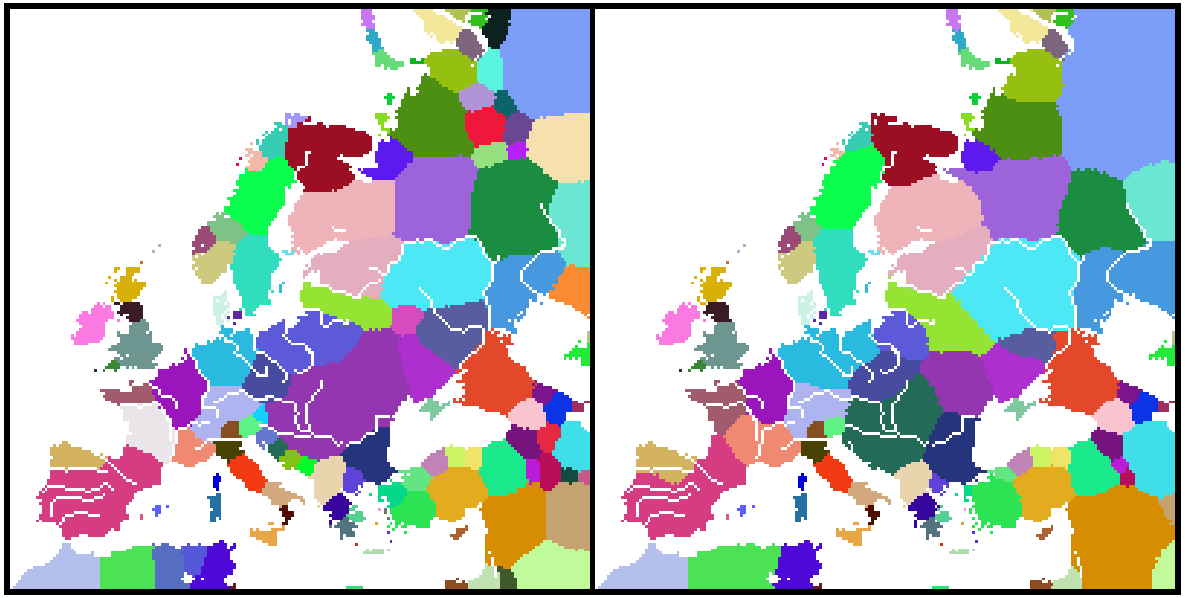
Think & Drink



*Joint with Frederik Ravn Klausen, QMATH, University of Copenhagen

The model

- Countries/Cultures/Languages/Ideas/... "fight" for territory
- Naively: strength of country $\sim \frac{\text{area}}{\text{perimeter}}$ has dimension of length.
- $\rightsquigarrow$ uninteresting dynamic of one large country. Instead

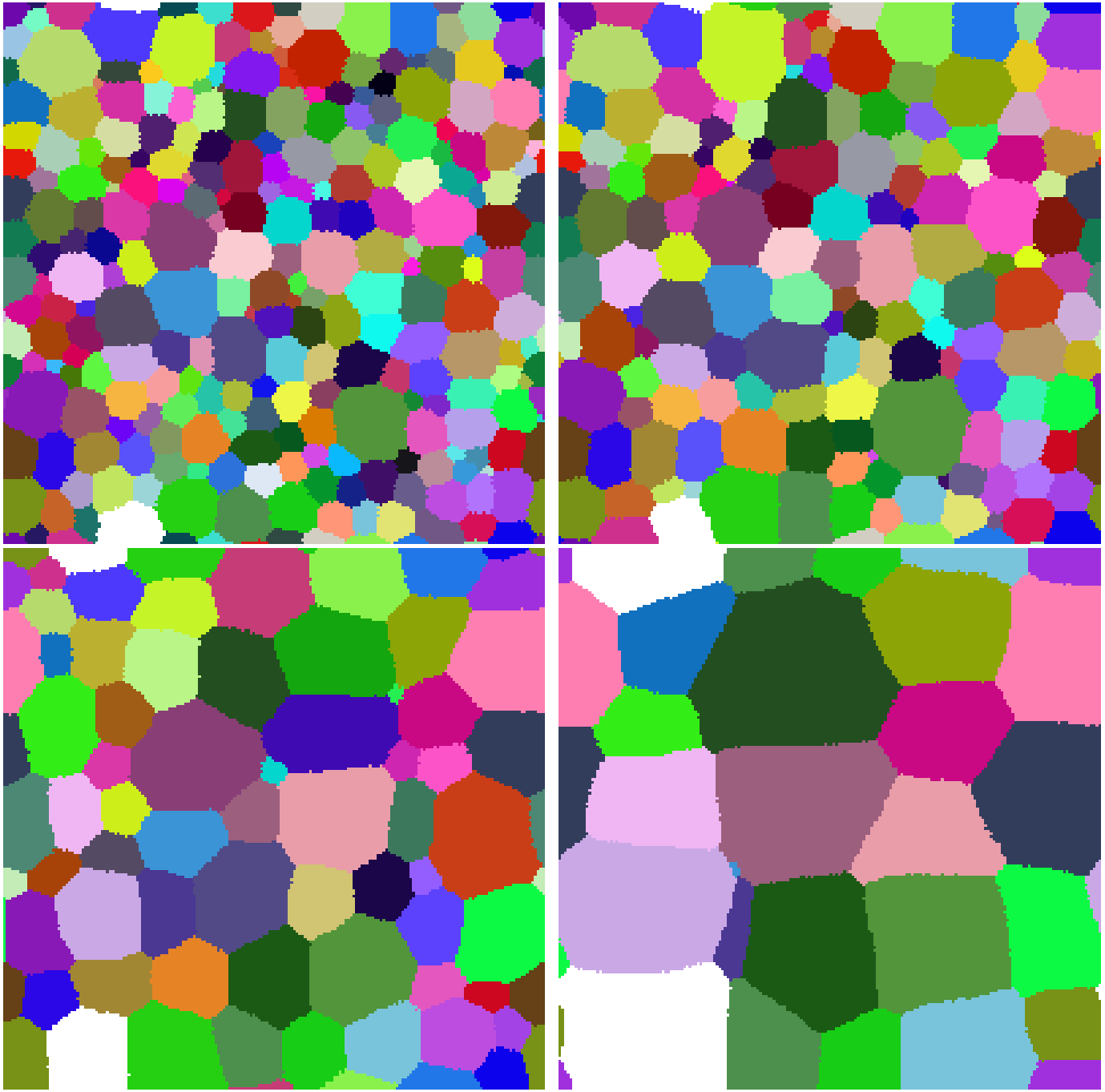
$$\text{strength} := \frac{\sqrt{\text{area}}}{\text{perimeter}}$$

- Dimensionless $\rightsquigarrow$ coexistence of countries of different sizes.
- Local dynamics "fighting" for each pixel - strength modulated by number of close neighbours belonging to country:

A	A	B	B	B
A	A	B	B	B
A	A	B	B	B
A	A	C	C	C
A	A	C	C	C

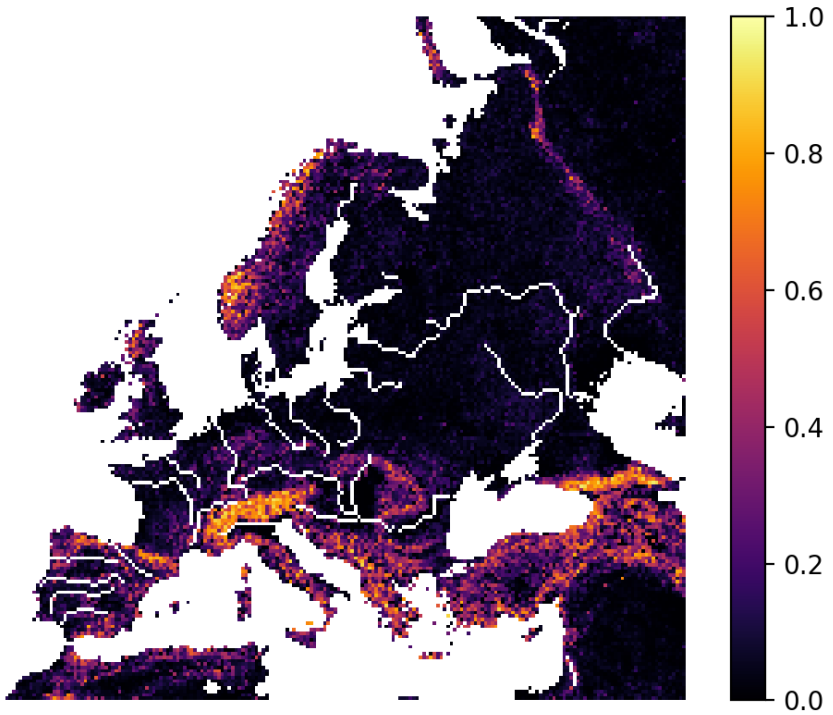
Neighbours: $8 \times$ A, $8 \times$ B, $5 \times$ C

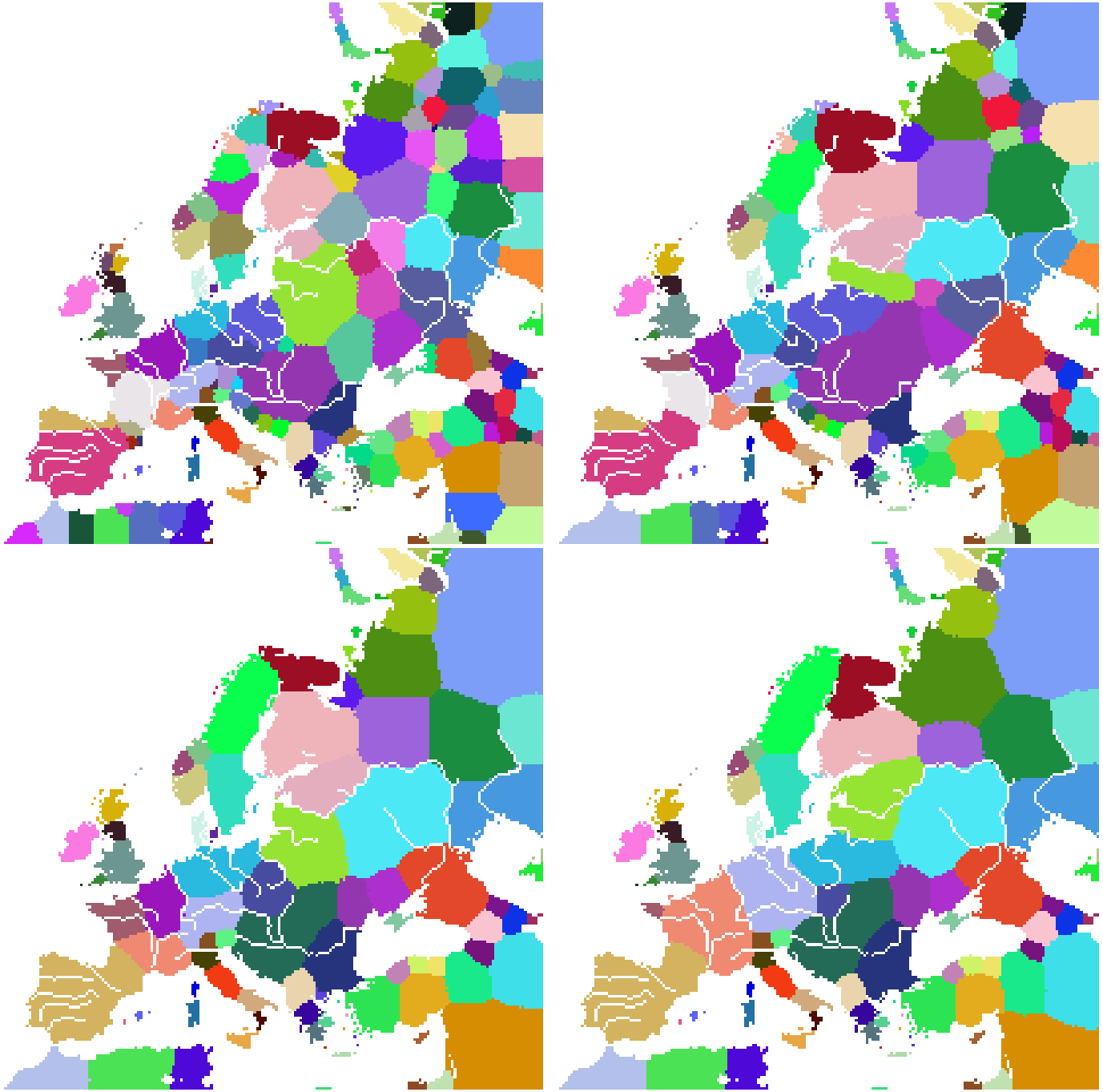
- Modulated by randomness $\rightsquigarrow$ parameter p, the **fluctuation**

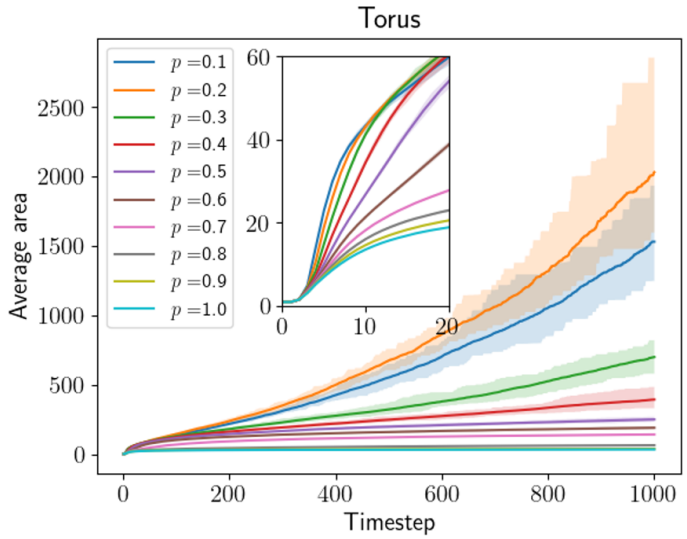


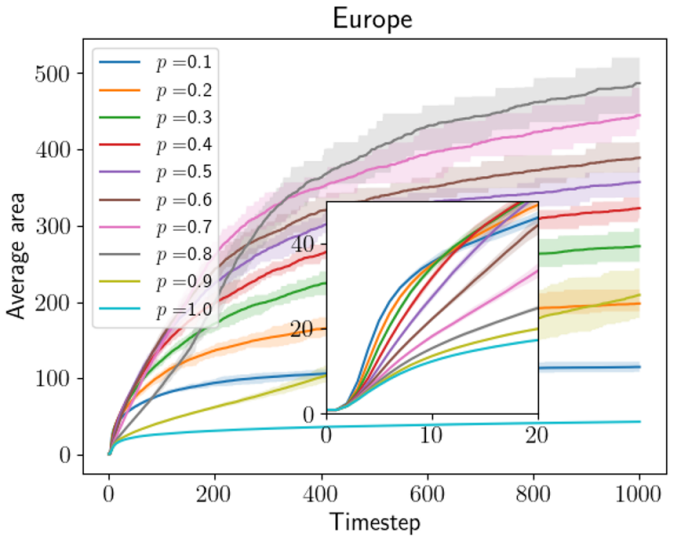
Geography

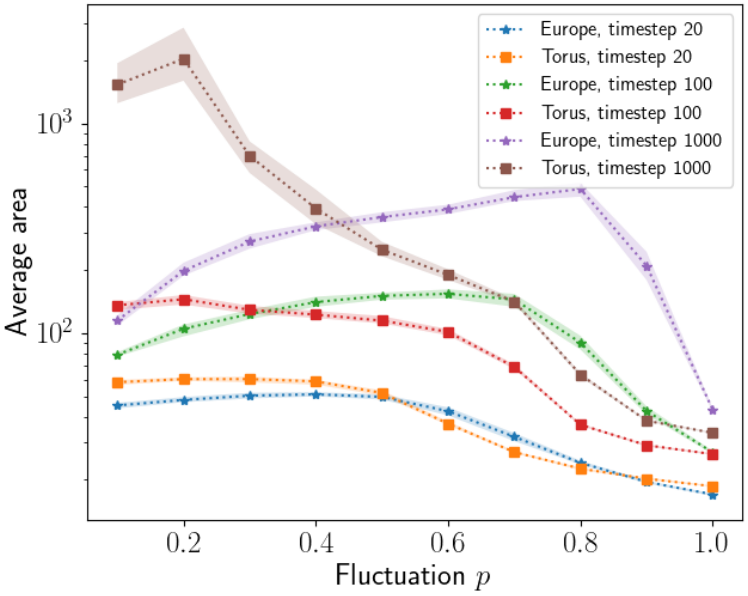
- Mountains
  1. Easier to "defend": get bonus for country already present - local effect. Modelled by parameter $D_m(= 2)$.
  2. Decrease to perimeter - global effect. Modelled by parameter $P_m(= 0.5)$.
- Sea/boundary - decrease to perimeter, only contributes $s = 0.5$.
- Rivers
  1. "$\sim$ sea" - decrease to perimeter, only contributes $s = 0.5$.
  2. Trade routes - if controlling both sides of river - Area bonus $A_r(= 8)$.



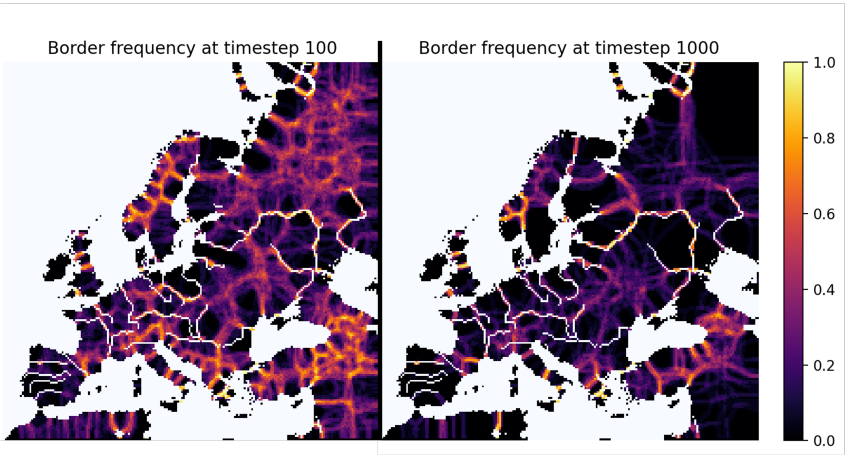
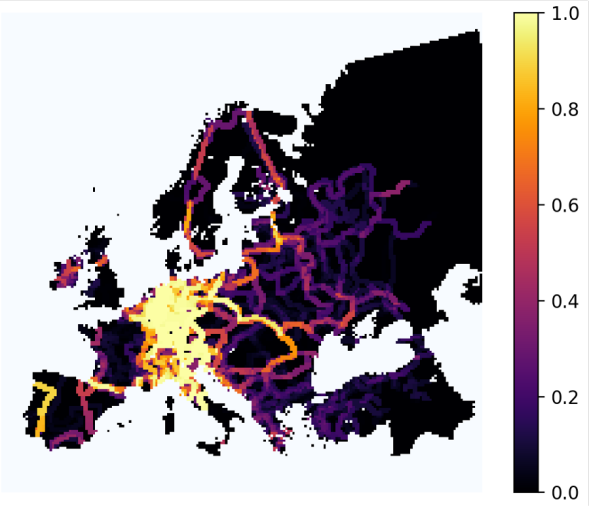
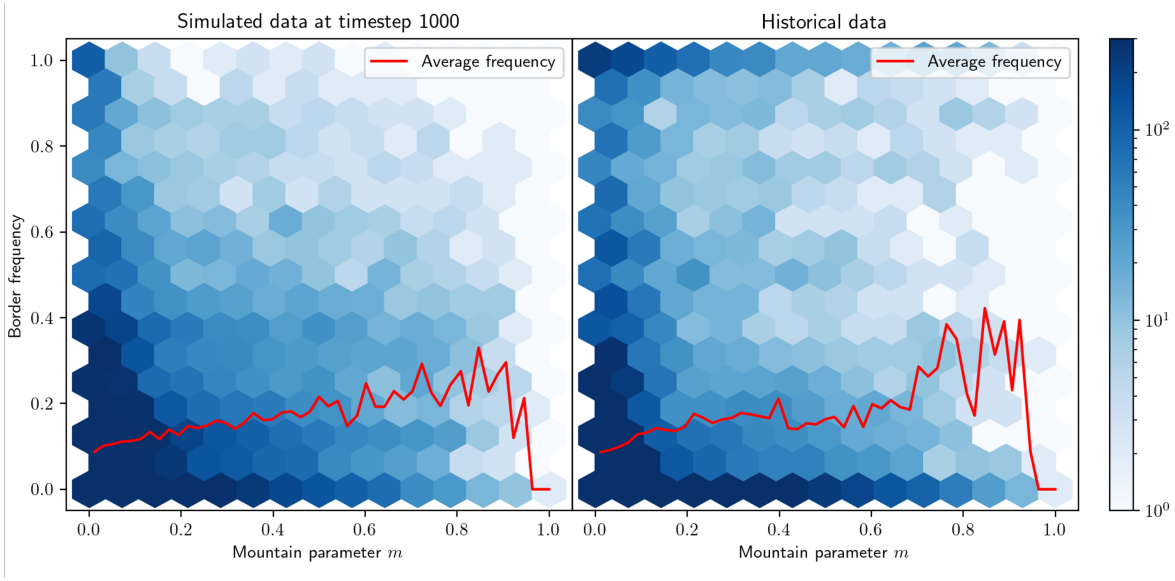








$\rightsquigarrow$ p is not an effective temperature

Comparison to historical data

Historical data: average borders between 1200-1790